

GWLabCh5 Solutions

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In this lab we're going to focus on how to use the `dplyr` package, another core member of the `tidyverse`. We'll illustrate the key ideas using data from the `nycflights13` package, and use `ggplot2` to help us understand the data.

Please read Chapter 5, edition 1, of *R for Data Science (GW)*, Garrett Golemund, Hadley Wickham, and complete the exercises below after you finish each section. Edit the markdown file which came with this html directly. Make sure to enter your R code in the chunks following each question to demonstrate your answers. Answers without demonstration will be given little credit. The function `glimpse()` is used to show only the graded portion of your answer.

For this lab, you will need to install the `nycflights13` data set. Go to your *Console* window, and type `install.packages("nycflight13")` before you run the `library()` call below.

```
library(nycflights13)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.3      v readr      2.1.4
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.4.3      v tibble    3.2.1
## v lubridate  1.9.2      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
glimpse(flights)
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
## $ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
## $ dep_delay  <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -1~
## $ arr_time   <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849, ~
## $ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851, ~
## $ arr_delay  <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -1~
## $ carrier    <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
## $ flight     <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
## $ tailnum    <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
## $ origin     <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA", ~
```

```
## $ dest      <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD", ~
## $ air_time  <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
## $ distance  <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
## $ hour      <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
## $ minute    <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
## $ time_hour <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-01 0~
```

5.2.4 Exercises

1.1 Using `filter()` then `glimpse()`, find all flights that had an arrival delay of two or more hours.

```
flights %>%
  filter(arr_delay >= 120) %>%
  glimpse()
```

```
## Rows: 10,200
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 811, 848, 957, 1114, 1505, 1525, 1549, 1558, 1732, 1803~
## $ sched_dep_time <int> 630, 1835, 733, 900, 1310, 1340, 1445, 1359, 1630, 1620~
## $ dep_delay <dbl> 101, 853, 144, 134, 115, 105, 64, 119, 62, 103, 290, 26~
## $ arr_time  <int> 1047, 1001, 1056, 1447, 1638, 1831, 1912, 1718, 2028, 2~
## $ sched_arr_time <int> 830, 1950, 853, 1222, 1431, 1626, 1656, 1515, 1825, 175~
## $ arr_delay <dbl> 137, 851, 123, 145, 127, 125, 136, 123, 123, 138, 338, ~
## $ carrier   <chr> "MQ", "MQ", "UA", "UA", "EV", "B6", "EV", "EV", "EV", "~
## $ flight    <int> 4576, 3944, 856, 1086, 4497, 525, 4181, 5712, 4092, 462~
## $ tailnum   <chr> "N531MQ", "N942MQ", "N534UA", "N76502", "N17984", "N231~
## $ origin    <chr> "LGA", "JFK", "EWR", "LGA", "EWR", "EWR", "EWR", "JFK", ~
## $ dest      <chr> "CLT", "BWI", "BOS", "IAH", "RIC", "MCO", "MCI", "IAD", ~
## $ air_time  <dbl> 118, 41, 37, 248, 63, 152, 234, 53, 119, 154, 213, 46, ~
## $ distance  <dbl> 544, 184, 200, 1416, 277, 937, 1092, 228, 533, 764, 113~
## $ hour      <dbl> 6, 18, 7, 9, 13, 13, 14, 13, 16, 16, 13, 14, 16, 17, 17~
## $ minute    <dbl> 30, 35, 33, 0, 10, 40, 45, 59, 30, 20, 25, 22, 45, 25, ~
## $ time_hour <dtm> 2013-01-01 06:00:00, 2013-01-01 18:00:00, 2013-01-01 0~
```

1.2 Using `filter()` then `glimpse()`, find all flights that flew to Houston (IAH or HOU)

```
flights %>%
  filter(dest == "IAH" | dest == "HOU") %>%
  glimpse()
```

```
## Rows: 9,313
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 517, 533, 623, 728, 739, 908, 1028, 1044, 1114, 1205, 1~
## $ sched_dep_time <int> 515, 529, 627, 732, 739, 908, 1026, 1045, 900, 1200, 11~
## $ dep_delay <dbl> 2, 4, -4, -4, 0, 0, 2, -1, 134, 5, 10, -2, 6, 6, 0, 12, ~
## $ arr_time  <int> 830, 850, 933, 1041, 1104, 1228, 1350, 1352, 1447, 1503~
```

```
## $ sched_arr_time <int> 819, 830, 932, 1038, 1038, 1219, 1339, 1351, 1222, 1505~
## $ arr_delay      <dbl> 11, 20, 1, 3, 26, 9, 11, 1, 145, -2, 38, 0, 12, 19, -9,~
## $ carrier        <chr> "UA", "UA", "UA", "UA", "UA", "UA", "UA", "UA", "UA", "~
## $ flight         <int> 1545, 1714, 496, 473, 1479, 1220, 1004, 455, 1086, 1461~
## $ tailnum        <chr> "N14228", "N24211", "N459UA", "N488UA", "N37408", "N122~
## $ origin         <chr> "EWR", "LGA", "LGA", "LGA", "EWR", "EWR", "LGA", "EWR",~
## $ dest           <chr> "IAH", "IAH", "IAH", "IAH", "IAH", "IAH", "IAH", "IAH",~
## $ air_time       <dbl> 227, 227, 229, 238, 249, 233, 237, 229, 248, 221, 253, ~
## $ distance       <dbl> 1400, 1416, 1416, 1416, 1400, 1400, 1416, 1400, 1416, 1~
## $ hour           <dbl> 5, 5, 6, 7, 7, 9, 10, 10, 9, 12, 11, 12, 13, 13, 14, 15~
## $ minute         <dbl> 15, 29, 27, 32, 39, 8, 26, 45, 0, 0, 58, 50, 0, 50, 30,~
## $ time_hour      <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-01 0~
```

1.3 Using `filter()` then `glimpse()`, were operated by United, American, or Delta.

```
flights %>%
  filter(carrier=="UA" | carrier=="AA" | carrier=="DL") %>%
  glimpse()
```

```
## Rows: 139,504
## Columns: 19
## $ year          <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month         <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day           <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time      <int> 517, 533, 542, 554, 554, 558, 558, 558, 559, 559, 602, ~
## $ sched_dep_time <int> 515, 529, 540, 600, 558, 600, 600, 600, 600, 600, 610, ~
## $ dep_delay     <dbl> 2, 4, 2, -6, -4, -2, -2, -2, -1, -1, -8, -4, -4, 0, 11,~
## $ arr_time      <int> 830, 850, 923, 812, 740, 753, 924, 923, 941, 854, 812, ~
## $ sched_arr_time <int> 819, 830, 850, 837, 728, 745, 917, 937, 910, 902, 820, ~
## $ arr_delay     <dbl> 11, 20, 33, -25, 12, 8, 7, -14, 31, -8, -8, -12, -8, -1~
## $ carrier       <chr> "UA", "UA", "AA", "DL", "UA", "AA", "UA", "UA", "AA", "~
## $ flight        <int> 1545, 1714, 1141, 461, 1696, 301, 194, 1124, 707, 1187,~
## $ tailnum       <chr> "N14228", "N24211", "N619AA", "N668DN", "N39463", "N3AL~
## $ origin        <chr> "EWR", "LGA", "JFK", "LGA", "EWR", "LGA", "JFK", "EWR",~
## $ dest          <chr> "IAH", "IAH", "MIA", "ATL", "ORD", "ORD", "LAX", "SFO",~
## $ air_time      <dbl> 227, 227, 160, 116, 150, 138, 345, 361, 257, 337, 170, ~
## $ distance      <dbl> 1400, 1416, 1089, 762, 719, 733, 2475, 2565, 1389, 2227~
## $ hour          <dbl> 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6~
## $ minute        <dbl> 15, 29, 40, 0, 58, 0, 0, 0, 0, 0, 10, 10, 10, 7, 0, 15,~
## $ time_hour     <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-01 0~
```

1.4 Using `filter()` and `glimpse()`, find all flights that departed in summer (July, August, and September). Use the `%in%` operator.

```
flights %>%
  filter(month %in% 7:9) %>%
  glimpse()
```

```
## Rows: 86,326
## Columns: 19
## $ year          <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month         <int> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7~
```

```
## $ day          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time     <int> 1, 2, 29, 43, 44, 46, 48, 58, 100, 100, 107, 111, 114, ~
## $ sched_dep_time <int> 2029, 2359, 2245, 2130, 2150, 2051, 2001, 2155, 2146, 2~
## $ dep_delay    <dbl> 212, 3, 104, 193, 174, 235, 287, 183, 194, 135, 142, 72~
## $ arr_time     <int> 236, 344, 151, 322, 300, 304, 308, 335, 327, 337, 158, ~
## $ sched_arr_time <int> 2359, 344, 1, 14, 100, 2358, 2305, 43, 30, 135, 2359, 3~
## $ arr_delay    <dbl> 157, 0, 110, 188, 120, 186, 243, 172, 177, 122, 119, 68~
## $ carrier      <chr> "B6", "B6", "B6", "B6", "AA", "B6", "VX", "B6", "B6", "~
## $ flight       <int> 915, 1503, 234, 1371, 185, 165, 415, 425, 1183, 623, 18~
## $ tailnum      <chr> "N653JB", "N805JB", "N348JB", "N794JB", "N324AA", "N640~
## $ origin       <chr> "JFK", "JFK", "JFK", "LGA", "JFK", "JFK", "JFK", "JFK", ~
## $ dest         <chr> "SFO", "SJU", "BTV", "FLL", "LAX", "PDX", "LAX", "TPA", ~
## $ air_time     <dbl> 315, 200, 66, 143, 297, 304, 298, 140, 126, 304, 39, 19~
## $ distance     <dbl> 2586, 1598, 266, 1076, 2475, 2454, 2475, 1005, 944, 247~
## $ hour         <dbl> 20, 23, 22, 21, 21, 20, 20, 21, 21, 22, 22, 23, 21, 5, ~
## $ minute       <dbl> 29, 59, 45, 30, 50, 51, 1, 55, 46, 45, 45, 59, 35, 0, 4~
## $ time_hour    <dtm> 2013-07-01 20:00:00, 2013-07-01 23:00:00, 2013-07-01 2~
```

1.5 Using `filter()` and `glimpse()`, find all flights that arrived more than two hours late, but didn't leave late.

```
flights %>%
  filter(arr_delay > 120, dep_delay <= 0) %>%
  glimpse()
```

```
## Rows: 29
## Columns: 19
## $ year          <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month         <int> 1, 10, 10, 10, 11, 3, 4, 4, 4, 5, 5, 6, 6, 6, 6, 6, 7, ~
## $ day           <int> 27, 7, 7, 16, 1, 18, 17, 18, 18, 22, 23, 5, 14, 24, 27, ~
## $ dep_time      <int> 1419, 1350, 1357, 657, 658, 1844, 1635, 558, 655, 1827, ~
## $ sched_dep_time <int> 1420, 1350, 1359, 700, 700, 1847, 1640, 600, 700, 1830, ~
## $ dep_delay     <dbl> -1, 0, -2, -3, -2, -3, -5, -2, -5, -3, 0, -11, -2, -3, ~
## $ arr_time      <int> 1754, 1736, 1858, 1258, 1329, 39, 2049, 1149, 1213, 221~
## $ sched_arr_time <int> 1550, 1526, 1654, 1056, 1015, 2219, 1845, 850, 950, 201~
## $ arr_delay     <dbl> 124, 130, 124, 122, 194, 140, 124, 179, 143, 127, 128, ~
## $ carrier       <chr> "MQ", "EV", "AA", "B6", "VX", "UA", "MQ", "AA", "AA", "~
## $ flight        <int> 3728, 5181, 1151, 3, 399, 389, 4540, 707, 2083, 4674, 4~
## $ tailnum       <chr> "N1EAMQ", "N611QX", "N3CMAA", "N703JB", "N629VA", "N560~
## $ origin        <chr> "EWR", "LGA", "LGA", "JFK", "JFK", "JFK", "LGA", "LGA", ~
## $ dest          <chr> "ORD", "MSN", "DFW", "SJU", "LAX", "SFO", "DTW", "DFW", ~
## $ air_time      <dbl> 135, 117, 192, 225, 336, 386, 130, 234, 230, 90, 82, 15~
## $ distance      <dbl> 719, 812, 1389, 1598, 2475, 2586, 502, 1389, 1372, 419, ~
## $ hour          <dbl> 14, 13, 13, 7, 7, 18, 16, 6, 7, 18, 18, 16, 17, 16, 21, ~
## $ minute        <dbl> 20, 50, 59, 0, 0, 47, 40, 0, 0, 30, 10, 15, 10, 5, 0, 2~
## $ time_hour     <dtm> 2013-01-27 14:00:00, 2013-10-07 13:00:00, 2013-10-07 1~
```

1.6 Using `filter()` and `glimpse()`, find all flights that were delayed by at least an hour, but made up over 30 minutes in flight

```
flights %>%
  filter(dep_delay >= 60, dep_delay - arr_delay > 30) %>%
  glimpse()
```

```
## Rows: 1,844
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 6, 7, 7~
## $ dep_time  <int> 2205, 2326, 1503, 1839, 1850, 1941, 1950, 2015, 2257, 1~
## $ sched_dep_time <int> 1720, 2130, 1221, 1700, 1745, 1759, 1845, 1915, 2000, 1~
## $ dep_delay <dbl> 285, 116, 162, 99, 65, 102, 65, 60, 177, 137, 145, 96, ~
## $ arr_time  <int> 46, 131, 1803, 2056, 2148, 2246, 2228, 2135, 45, 2135, ~
## $ sched_arr_time <int> 2040, 18, 1555, 1950, 2120, 2139, 2227, 2111, 2224, 195~
## $ arr_delay <dbl> 246, 73, 128, 66, 28, 67, 1, 24, 141, 105, 97, 55, 172,~
## $ carrier   <chr> "AA", "B6", "UA", "AA", "AA", "UA", "B6", "9E", "9E", "~
## $ flight    <int> 1999, 199, 551, 575, 177, 979, 91, 3525, 3439, 575, 177~
## $ tailnum   <chr> "N5DNAA", "N594JB", "N835UA", "N631AA", "N332AA", "N402~
## $ origin    <chr> "EWR", "JFK", "EWR", "JFK", "JFK", "EWR", "JFK", "JFK",~
## $ dest      <chr> "MIA", "LAS", "SFO", "EGE", "SFO", "PHX", "OAK", "ORD",~
## $ air_time  <dbl> 146, 290, 320, 239, 314, 274, 319, 110, 90, 239, 327, 2~
## $ distance  <dbl> 1085, 2248, 2565, 1747, 2586, 2133, 2576, 740, 589, 174~
## $ hour      <dbl> 17, 21, 12, 17, 17, 17, 18, 19, 20, 17, 17, 19, 17, 19,~
## $ minute    <dbl> 20, 30, 21, 0, 45, 59, 45, 15, 0, 0, 45, 12, 30, 20, 58~
## $ time_hour <dtm> 2013-01-01 17:00:00, 2013-01-01 21:00:00, 2013-01-03 1~
```

1.7 Using `filter()` and `glimpse()`, find all flights that departed between midnight and 6am (inclusive)

```
# in this dataset midnight is represented by 2400 (for reference)
flights %>%
  filter(dep_time<=600 | dep_time==2400) %>%
  glimpse()
```

```
## Rows: 9,373
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
## $ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
## $ dep_delay <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -1~
## $ arr_time  <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,~
## $ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,~
## $ arr_delay <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -1~
## $ carrier   <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
## $ flight    <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
## $ tailnum   <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
## $ origin    <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA",~
## $ dest      <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD",~
## $ air_time  <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
## $ distance  <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
## $ hour      <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
## $ minute    <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
## $ time_hour <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-01 0~
```

2.0 `between(x,left,right)` returns a logical vector indicating those elements of `x` which satisfy `left<= x <= right`. Here are two examples:

```
# Example 1: Find flights departing between 6:00 and 10:00 am
flights %>%
  filter(between(dep_time, 600, 1000)) %>%
  glimpse()
```

```
## Rows: 89,820
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 600, 600, 601, 602, 602, 606, 606, 607, 608, 611, 613, ~
## $ sched_dep_time <int> 600, 600, 600, 610, 605, 610, 610, 607, 600, 600, 610, ~
## $ dep_delay <dbl> 0, 0, 1, -8, -3, -4, -4, 0, 8, 11, 3, 0, 0, -8, 13, -4,~
## $ arr_time  <int> 851, 837, 844, 812, 821, 858, 837, 858, 807, 945, 925, ~
## $ sched_arr_time <int> 858, 825, 850, 820, 805, 910, 845, 915, 735, 931, 921, ~
## $ arr_delay <dbl> -7, 12, -6, -8, 16, -12, -8, -17, 32, 14, 4, -21, -9, 3~
## $ carrier   <chr> "B6", "MQ", "B6", "DL", "MQ", "AA", "DL", "UA", "MQ", "~
## $ flight    <int> 371, 4650, 343, 1919, 4401, 1895, 1743, 1077, 3768, 303~
## $ tailnum   <chr> "N595JB", "N542MQ", "N644JB", "N971DL", "N730MQ", "N633~
## $ origin    <chr> "LGA", "LGA", "EWR", "LGA", "LGA", "EWR", "JFK", "EWR",~
## $ dest      <chr> "FLL", "ATL", "PBI", "MSP", "DTW", "MIA", "ATL", "MIA",~
## $ air_time  <dbl> 152, 134, 147, 170, 105, 152, 128, 157, 139, 366, 175, ~
## $ distance  <dbl> 1076, 762, 1023, 1020, 502, 1085, 760, 1085, 719, 2586,~
## $ hour      <dbl> 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6~
## $ minute    <dbl> 0, 0, 0, 10, 5, 10, 10, 7, 0, 0, 10, 15, 15, 30, 10, 27~
## $ time_hour <dtm> 2013-01-01 06:00:00, 2013-01-01 06:00:00, 2013-01-01 0~
```

```
# Example 2: Find flights with arrival delay between 30 and 60 minutes
flights %>%
  filter(between(arr_delay, 30, 60)) %>%
  glimpse()
```

```
## Rows: 25,013
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ dep_time  <int> 542, 559, 608, 635, 702, 724, 732, 749, 754, 826, 833, ~
## $ sched_dep_time <int> 540, 600, 600, 635, 700, 730, 645, 710, 755, 715, 835, ~
## $ dep_delay <dbl> 2, -1, 8, 0, 2, -6, 47, 39, -1, 71, -2, 43, 0, 4, 29, 3~
## $ arr_time  <int> 923, 941, 807, 1028, 1058, 1111, 1011, 939, 1103, 1136,~
## $ sched_arr_time <int> 850, 910, 735, 940, 1014, 1040, 941, 850, 1030, 1045, 1~
## $ arr_delay <dbl> 33, 31, 32, 48, 44, 31, 30, 49, 33, 51, 32, 50, 40, 30,~
## $ carrier   <chr> "AA", "AA", "MQ", "AA", "B6", "AA", "UA", "MQ", "WN", "~
## $ flight    <int> 1141, 707, 3768, 711, 671, 715, 1111, 3737, 733, 443, 8~
## $ tailnum   <chr> "N619AA", "N3DUAA", "N9EAMQ", "N3GKAA", "N779JB", "N541~
## $ origin    <chr> "JFK", "LGA", "EWR", "LGA", "JFK", "LGA", "EWR", "EWR",~
## $ dest      <chr> "MIA", "DFW", "ORD", "DFW", "LAX", "DFW", "MCO", "ORD",~
## $ air_time  <dbl> 160, 257, 139, 248, 381, 254, 145, 148, 279, 160, 257, ~
## $ distance  <dbl> 1089, 1389, 719, 1389, 2475, 1389, 937, 719, 1620, 1089~
## $ hour      <dbl> 5, 6, 6, 6, 7, 7, 6, 7, 7, 7, 8, 8, 9, 9, 9, 9, 10, ~
## $ minute    <dbl> 40, 0, 0, 35, 0, 30, 45, 10, 55, 15, 35, 20, 5, 25, 4, ~
## $ time_hour <dtm> 2013-01-01 05:00:00, 2013-01-01 06:00:00, 2013-01-01 0~
```

Here is the code rewritten for 1.4

```
flights %>%
  filter(between(month,7,9)) %>%
  glimpse()
```

Here is the code rewritten for 1.7

```
flights %>%
  filter(between(dep_time, 0, 600) | dep_time == 2400) %>%
  glimpse()
```

3.0 How many flights have a missing dep_time? What other variables are missing? What might these rows represent?

```
summary(flights)
```

```
##      year      month      day      dep_time      sched_dep_time
##  Min.   :2013      Min.   : 1.000      Min.   : 1.00      Min.    : 1      Min.    : 106
##  1st Qu.:2013      1st Qu.: 4.000      1st Qu.: 8.00      1st Qu.: 907      1st Qu.: 906
##  Median :2013      Median : 7.000      Median :16.00      Median :1401      Median :1359
##  Mean   :2013      Mean   : 6.549      Mean   :15.71      Mean   :1349      Mean   :1344
##  3rd Qu.:2013      3rd Qu.:10.000     3rd Qu.:23.00      3rd Qu.:1744      3rd Qu.:1729
##  Max.   :2013      Max.   :12.000     Max.   :31.00      Max.   :2400      Max.   :2359
##
##                                     NA's      :8255
##  dep_delay      arr_time      sched_arr_time      arr_delay
##  Min.    : -43.00      Min.    : 1      Min.    : 1      Min.    : -86.000
##  1st Qu.: -5.00      1st Qu.:1104      1st Qu.:1124      1st Qu.: -17.000
##  Median : -2.00      Median :1535      Median :1556      Median : -5.000
##  Mean    : 12.64      Mean    :1502      Mean    :1536      Mean    : 6.895
##  3rd Qu.: 11.00      3rd Qu.:1940      3rd Qu.:1945      3rd Qu.: 14.000
##  Max.    :1301.00      Max.    :2400      Max.    :2359      Max.    :1272.000
##  NA's    :8255      NA's    :8713      NA's    :9430
##  carrier      flight      tailnum      origin
##  Length:336776      Min.    : 1      Length:336776      Length:336776
##  Class :character      1st Qu.: 553      Class :character      Class :character
##  Mode  :character      Median :1496      Mode  :character      Mode  :character
##                                     Mean    :1972
##                                     3rd Qu.:3465
##                                     Max.    :8500
##
##  dest      air_time      distance      hour
##  Length:336776      Min.    : 20.0      Min.    : 17      Min.    : 1.00
##  Class :character      1st Qu.: 82.0      1st Qu.: 502      1st Qu.: 9.00
##  Mode  :character      Median :129.0      Median : 872      Median :13.00
##                                     Mean    :150.7      Mean    :1040      Mean    :13.18
##                                     3rd Qu.:192.0      3rd Qu.:1389      3rd Qu.:17.00
##                                     Max.    :695.0      Max.    :4983      Max.    :23.00
##                                     NA's    :9430
##  minute      time_hour
##  Min.    : 0.00      Min.    :2013-01-01 05:00:00.00
##  1st Qu.: 8.00      1st Qu.:2013-04-04 13:00:00.00
##  Median :29.00      Median :2013-07-03 10:00:00.00
```

```
## Mean      :26.23      Mean      :2013-07-03 05:22:54.64
## 3rd Qu.   :44.00      3rd Qu.   :2013-10-01 07:00:00.00
## Max.      :59.00      Max.      :2013-12-31 23:00:00.00
##
```

The summary shows us that 8255 flights have a missing dep_time, 8255 have a missing dep_delay, 8713 have a missing arr_time, 9430 have a missing arr_delay, and 9430 have a missing air_time. These can be flights that failed to depart or arrive (maybe cancelled flights). Or they can just be lost data about the flights.

4.0 Why is `NA ^ 0` not missing? Why is `NA | TRUE` not missing? Why is `FALSE & NA` not missing? Can you figure out the general rule? (`NA * 0` is a tricky counterexample!)

```
NA ^ 0
```

```
## [1] 1
```

```
NA | TRUE
```

```
## [1] TRUE
```

```
FALSE & NA
```

```
## [1] FALSE
```

```
NA * 0
```

```
## [1] NA
```

`NA ^ 0` Not missing because anything to the power 0 returns 1. `NA | TRUE` Not missing because `|` evaluates to TRUE if any of the terms is true. `FALSE & NA` Not missing because `&` evaluates to TRUE ONLY if both terms are TRUE, which is not the case here. `NA * 0` is not 0 because it is undefined. it is like doing `0 * infinity` and `0 * -infinity`

5.3.1 Exercises

1.0 How could you use `arrange()` to sort all instances in `flights` have missing values of `dep_time` to the start? (Hint: use `is.na()`). Please use `glimpse()`.

```
flights %>%
  arrange(desc(is.na(dep_time))) %>%
  glimpse()
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3~
## $ dep_time  <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ sched_dep_time <int> 1630, 1935, 1500, 600, 1540, 1620, 1355, 1420, 1321, 15~
```



```
## $ dep_delay      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ arr_time       <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ sched_arr_time <int> 1815, 2240, 1825, 901, 1747, 1746, 1459, 1644, 1536, 19~
## $ arr_delay      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ carrier        <chr> "EV", "AA", "AA", "B6", "EV", "EV", "EV", "EV", "EV", "~
## $ flight         <int> 4308, 791, 1925, 125, 4352, 4406, 4434, 4935, 3849, 133~
## $ tailnum        <chr> "N18120", "N3EHAA", "N3EVAA", "N618JB", "N10575", "N139~
## $ origin         <chr> "EWR", "LGA", "LGA", "JFK", "EWR", "EWR", "EWR", "EWR", ~
## $ dest           <chr> "RDU", "DFW", "MIA", "FLL", "CVG", "PIT", "MHT", "ATL", ~
## $ air_time       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ distance       <dbl> 416, 1389, 1096, 1069, 569, 319, 209, 746, 645, 2475, 1~
## $ hour           <dbl> 16, 19, 15, 6, 15, 16, 13, 14, 13, 15, 13, 16, 6, 10, 1~
## $ minute         <dbl> 30, 35, 0, 0, 40, 20, 55, 20, 21, 45, 30, 1, 45, 30, 25~
## $ time_hour      <dtm> 2013-01-01 16:00:00, 2013-01-01 19:00:00, 2013-01-01 1~
```

```
flights %>%  
  arrange(desc(dep_delay)) %>%  
  glimpse()
```

Here the most delayed flight for example is flight 51 (carrier: HA) which was scheduled to depart January 9 2013 with delay of 1301 minutes.

```
## Rows: 336,776
```

```
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 12, 2, 11, 1, 1, 8, 10, 3, 3, 5, 5, 9, 10, 11, 12, 5, 9~
## $ day       <int> 7, 3, 10, 11, 29, 9, 23, 30, 2, 5, 14, 18, 29, 18, 25, ~
## $ dep_time  <int> 2040, 2022, 1408, 1900, 1703, 729, 1907, 2030, 1431, 93~
## $ sched_dep_time <int> 2123, 2055, 1440, 1930, 1730, 755, 1932, 2055, 1455, 95~
## $ dep_delay <dbl> -43, -33, -32, -30, -27, -26, -25, -25, -24, -24, -24, ~
## $ arr_time  <int> 40, 2240, 1549, 2233, 1947, 1002, 2143, 2213, 1601, 122~
## $ sched_arr_time <int> 2352, 2338, 1559, 2243, 1957, 955, 2143, 2250, 1631, 13~
## $ arr_delay <dbl> 48, -58, -10, -10, -10, 7, 0, -37, -30, -44, -21, -33, ~
## $ carrier   <chr> "B6", "DL", "EV", "DL", "F9", "MQ", "EV", "MQ", "9E", "~
## $ flight    <int> 97, 1715, 5713, 1435, 837, 3478, 4361, 4573, 3318, 375,~
## $ tailnum   <chr> "N592JB", "N612DL", "N825AS", "N934DL", "N208FR", "N711~
## $ origin    <chr> "JFK", "LGA", "LGA", "LGA", "LGA", "LGA", "EWR", "LGA",~
## $ dest      <chr> "DEN", "MSY", "IAD", "TPA", "DEN", "DTW", "TYS", "DTW",~
## $ air_time  <dbl> 265, 162, 52, 139, 250, 88, 111, 87, 55, 150, 107, 125,~
## $ distance  <dbl> 1626, 1183, 229, 1010, 1620, 502, 631, 502, 301, 1076, ~
## $ hour      <dbl> 21, 20, 14, 19, 17, 7, 19, 20, 14, 9, 9, 16, 21, 20, 20~
## $ minute    <dbl> 23, 55, 40, 30, 30, 55, 32, 55, 55, 58, 38, 55, 6, 29, ~
## $ time_hour <dtm> 2013-12-07 21:00:00, 2013-02-03 20:00:00, 2013-11-10 1~
```

We can see for example that flight 97 (carrier: B6) which was scheduled to depart December 7 2013 AT 21:23 department 43 minutes early (dep_delay=-43).

3. Sort flights to find the highest-speed flights. Please use `glimpse()`.

```
flights %>%
  arrange(desc(distance / air_time)) %>%
  glimpse()
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 5, 7, 5, 3, 1, 11, 2, 11, 11, 11, 11, 11, 12, 2, 2, 11,~
## $ day       <int> 25, 2, 13, 23, 12, 17, 21, 17, 16, 16, 17, 17, 5, 10, 1~
## $ dep_time  <int> 1709, 1558, 2040, 1914, 1559, 650, 2355, 759, 2003, 234~
## $ sched_dep_time <int> 1700, 1513, 2025, 1910, 1600, 655, 2358, 800, 1925, 235~
## $ dep_delay <dbl> 9, 45, 15, 4, -1, -5, -3, -1, 38, -10, -4, 1, -2, -2, 6~
## $ arr_time  <int> 1923, 1745, 2225, 2045, 1849, 1059, 412, 1212, 17, 402,~
## $ sched_arr_time <int> 1937, 1719, 2226, 2043, 1917, 1150, 438, 1255, 36, 440,~
## $ arr_delay <dbl> -14, 26, -1, 2, -28, -51, -26, -43, -19, -38, -38, -32,~
## $ carrier   <chr> "DL", "EV", "EV", "EV", "DL", "DL", "B6", "AA", "DL", "~
## $ flight    <int> 1499, 4667, 4292, 3805, 1902, 315, 707, 936, 347, 1503,~
## $ tailnum   <chr> "N666DN", "N17196", "N14568", "N12567", "N956DL", "N376~
## $ origin    <chr> "LGA", "EWR", "EWR", "EWR", "LGA", "JFK", "JFK", "JFK",~
## $ dest      <chr> "ATL", "MSP", "GSP", "BNA", "PBI", "SJU", "SJU", "STT",~
## $ air_time  <dbl> 65, 93, 55, 70, 105, 170, 172, 175, 173, 173, 173, 173,~
## $ distance  <dbl> 762, 1008, 594, 748, 1035, 1598, 1598, 1623, 1598, 1598~
## $ hour      <dbl> 17, 15, 20, 19, 16, 6, 23, 8, 19, 23, 8, 19, 19, 17, 18~
## $ minute    <dbl> 0, 13, 25, 10, 0, 55, 58, 0, 25, 59, 55, 25, 0, 0, 55, ~
## $ time_hour <dtm> 2013-05-25 17:00:00, 2013-07-02 15:00:00, 2013-05-13 2~
```

4.0 Which flights travelled the farthest? Which travelled the shortest? Please use `glimpse()`.

```
flights %>%
  arrange(desc(distance)) %>%
  glimpse()
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ dep_time  <int> 857, 909, 914, 900, 858, 1019, 1042, 901, 641, 859, 855~
## $ sched_dep_time <int> 900, 900, 900, 900, 900, 900, 900, 900, 900, 900, 900, ~
## $ dep_delay <dbl> -3, 9, 14, 0, -2, 79, 102, 1, 1301, -1, -5, 1, -4, -1, ~
## $ arr_time  <int> 1516, 1525, 1504, 1516, 1519, 1558, 1620, 1504, 1242, 1~
## $ sched_arr_time <int> 1530, 1530, 1530, 1530, 1530, 1530, 1530, 1530, 1530, 1~
## $ arr_delay <dbl> -14, -5, -26, -14, -11, 28, 50, -26, 1272, -41, -48, -3~
## $ carrier   <chr> "HA", "HA", "HA", "HA", "HA", "HA", "HA", "HA", "HA", "HA", "~
## $ flight    <int> 51, 51, 51, 51, 51, 51, 51, 51, 51, 51, 51, 51, 51, 51, ~
## $ tailnum   <chr> "N380HA", "N380HA", "N380HA", "N384HA", "N381HA", "N385~
## $ origin    <chr> "JFK", "JFK", "JFK", "JFK", "JFK", "JFK", "JFK", "JFK", "JFK", ~
## $ dest      <chr> "HNL", "HNL", "HNL", "HNL", "HNL", "HNL", "HNL", "HNL", "HNL", ~
## $ air_time  <dbl> 659, 638, 616, 639, 635, 611, 612, 645, 640, 633, 613, ~
## $ distance  <dbl> 4983, 4983, 4983, 4983, 4983, 4983, 4983, 4983, 4983, 4~
## $ hour      <dbl> 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9~
## $ minute    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ time_hour <dtm> 2013-01-01 09:00:00, 2013-01-02 09:00:00, 2013-01-03 0~
```

Here for example the flight that travelled the farthest (highest distance) is flight 51 (carrier: “HA”) which was scheduled to depart January 1 2013 from JFK to HNL.

```
flights %>%
  arrange(distance) %>%
  glimpse()
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
## $ month     <int> 7, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day       <int> 27, 3, 4, 4, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
## $ dep_time  <int> NA, 2127, 1240, 1829, 2128, 1155, 2125, 2124, 2127, 212~
## $ sched_dep_time <int> 106, 2129, 1200, 1615, 2129, 1200, 2129, 2129, 2130, 21~
## $ dep_delay <dbl> NA, -2, 40, 134, -1, -5, -4, -5, -3, -3, 4, 6, -4, 74, ~
## $ arr_time  <int> NA, 2222, 1333, 1937, 2218, 1241, 2224, 2212, 2304, 221~
## $ sched_arr_time <int> 245, 2224, 1306, 1721, 2224, 1306, 2224, 2224, 2225, 22~
## $ arr_delay <dbl> NA, -2, 27, 136, -6, -25, 0, -12, 39, -7, -1, 9, -14, 9~
## $ carrier   <chr> "US", "EV", "EV", "EV", "EV", "EV", "EV", "EV", "EV", "EV", "~
## $ flight    <int> 1632, 3833, 4193, 4502, 4645, 4193, 4619, 4619, 4619, 4~
## $ tailnum   <chr> NA, "N13989", "N14972", "N15983", "N27962", "N14902", "~
## $ origin    <chr> "EWR", "EWR", "EWR", "EWR", "EWR", "EWR", "EWR", "EWR", "EWR", ~
## $ dest      <chr> "LGA", "PHL", "PHL", "PHL", "PHL", "PHL", "PHL", "PHL", "PHL", ~
## $ air_time  <dbl> NA, 30, 30, 28, 32, 29, 22, 25, 30, 27, 30, 30, 36, 31, ~
## $ distance  <dbl> 17, 80, 80, 80, 80, 80, 80, 80, 80, 80, 80, 80, 80, 80, ~
## $ hour      <dbl> 1, 21, 12, 16, 21, 12, 21, 21, 21, 21, 21, 21, 16, 21, ~
```

```
## $ minute      <dbl> 6, 29, 0, 15, 29, 0, 29, 29, 30, 29, 29, 29, 17, 29, 29~
## $ time_hour   <dtm> 2013-07-27 01:00:00, 2013-01-03 21:00:00, 2013-01-04 1~
```

Here for example the flight that travelled the shortest distance is flight 51 (carrier: “US”) which was scheduled to depart July 27 2013 from EWR to LGA.

Section 5.4.1 Exercises

1. Brainstorm as many ways as possible to select `dep_time`, `dep_delay`, `arr_time`, and `arr_delay` from `flights`.

```
select(flights, dep_time, dep_delay, arr_time, arr_delay)
select(flights, c(dep_time, dep_delay, arr_time, arr_delay))
select(flights, "dep_time", "dep_delay", "arr_time", "arr_delay")
select(flights, 4, 6, 7, 9)
flights %>%
  select(dep_time, dep_delay, arr_time, arr_delay)
flights %>%
  select_("dep_time", "dep_delay", "arr_time", "arr_delay")
select(flights, any_of(c("dep_time", "dep_delay", "arr_time", "arr_delay")))
select(flights, starts_with("dep_"), starts_with("arr_"))
```

2. What happens if you include the name of a variable multiple times in a `select()` call? Why is this useful?

```
flights %>%
  select(dep_delay, dep_delay, dep_delay)
```

```
## # A tibble: 336,776 x 1
##   dep_delay
##   <dbl>
## 1         2
## 2         4
## 3         2
## 4        -1
## 5        -6
## 6        -4
## 7        -5
## 8        -3
## 9        -3
## 10       -2
## # i 336,766 more rows
```

Nothing happens when we include the variable multiple times inside `select()`. Duplicates are prevented. This is useful because we don't have to worry about them.

3. What do the `any_of()`, `all_of()` and `one_of()` functions do? Why might it be helpful in conjunction with this vector?

```
vars <- c("year", "month", "day", "dep_delay", "arr_delay")
```

```
vars <- c("year", "month", "day", "dep_delay", "arr_delay")
select(flights, any_of(vars))
```

```
## # A tibble: 336,776 x 5
##   year month   day dep_delay arr_delay
##   <int> <int> <int>     <dbl>     <dbl>
## 1  2013     1     1         2         11
## 2  2013     1     1         4         20
## 3  2013     1     1         2         33
## 4  2013     1     1        -1        -18
## 5  2013     1     1        -6        -25
## 6  2013     1     1        -4         12
## 7  2013     1     1        -5         19
## 8  2013     1     1        -3        -14
## 9  2013     1     1        -3         -8
## 10 2013     1     1        -2          8
## # i 336,766 more rows
```

```
select(flights, all_of(vars))
```

```
## # A tibble: 336,776 x 5
##   year month   day dep_delay arr_delay
##   <int> <int> <int>     <dbl>     <dbl>
## 1  2013     1     1         2         11
## 2  2013     1     1         4         20
## 3  2013     1     1         2         33
## 4  2013     1     1        -1        -18
## 5  2013     1     1        -6        -25
## 6  2013     1     1        -4         12
## 7  2013     1     1        -5         19
## 8  2013     1     1        -3        -14
## 9  2013     1     1        -3         -8
## 10 2013     1     1        -2          8
## # i 336,766 more rows
```

```
select(flights, one_of(vars))
```

```
## # A tibble: 336,776 x 5
##   year month   day dep_delay arr_delay
##   <int> <int> <int>     <dbl>     <dbl>
## 1  2013     1     1         2         11
## 2  2013     1     1         4         20
## 3  2013     1     1         2         33
## 4  2013     1     1        -1        -18
## 5  2013     1     1        -6        -25
## 6  2013     1     1        -4         12
## 7  2013     1     1        -5         19
## 8  2013     1     1        -3        -14
## 9  2013     1     1        -3         -8
## 10 2013     1     1        -2          8
## # i 336,766 more rows
```

These functions return all the variables we want, in this case all the variables in the vars vector. This is useful because it is easier to generate character vectors with variable names.

4. How do the `select` helpers deal with case by default? How can you change that default?

```
select(flights, contains("TIME"))
```

```
## # A tibble: 336,776 x 6
##   dep_time sched_dep_time arr_time sched_arr_time air_time time_hour
##   <int>      <int>      <int>      <int>      <dbl> <dtm>
## 1      517          515      830          819      227 2013-01-01 05:00:00
## 2      533          529      850          830      227 2013-01-01 05:00:00
## 3      542          540      923          850      160 2013-01-01 05:00:00
## 4      544          545     1004         1022      183 2013-01-01 05:00:00
## 5      554          600      812          837      116 2013-01-01 06:00:00
## 6      554          558      740          728      150 2013-01-01 05:00:00
## 7      555          600      913          854      158 2013-01-01 06:00:00
## 8      557          600      709          723       53 2013-01-01 06:00:00
## 9      557          600      838          846      140 2013-01-01 06:00:00
## 10     558          600      753          745      138 2013-01-01 06:00:00
## # i 336,766 more rows
```

```
select(flights, contains("TIME", ignore.case = FALSE))
```

```
## # A tibble: 336,776 x 0
```

By default `contains()` is not case sensitive. We can change that by using `ignore.case=FALSE`